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Paper Practice 1.2 • WORKED KEY

Algebraic Expressions

Show all of your work in the space provided. Define your variables when a problem asks you to. Circle your final answers.

1. Write a verbal expression for each algebraic expression. (Translate)

a. $7x^3 + 4$

Answer: seven times a number cubed, increased by four

b. $(3/4)n^2$

Answer: three-fourths of a number squared

c. $(m^5 + 2p) \div 9$

Answer: the sum of a number to the fifth power and twice a second number, all divided by nine

2. Write an algebraic expression for each verbal expression. (Translate)

a. three times the quantity k plus 8

Answer: $3(k + 8)$

b. seven less than a number squared

Answer: $n^2 - 7$ (not $7 - n^2$)

c. twenty increased by the product of five and d

Answer: $20 + 5d$

3. Evaluate $3x^2 - 2y$, if $x = 4$ and $y = 5$. (Evaluate)

Answer: $3(16) - 2(5) = 48 - 10 = 38$

4. Evaluate $b(12 - c)/a^2$, if $a = 2$, $b = 5$, and $c = 4$. (Evaluate)

Answer: $5(8) \div 4 = 40 \div 4 = 10$

5. Evaluate $a^3(4b - a + 1) - c$, if $a = 2$, $b = 3$, and $c = 6$. (Evaluate)

Answer: $8(12 - 2 + 1) - 6 = 8(11) - 6 = 88 - 6 = 82$

6. Evaluate $b(7 - c)/a^2$, if $a = 3$, $b = 9$, and $c = 3$. (Evaluate)

Answer: $9(4) \div 9 = 36 \div 9 = 4$

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7. Identify the terms, the coefficients, and the constant in $6x^2 + 4x - 9$. (Vocabulary)

Answer: Terms: $6x^2$, $4x$, -9. Coefficients: 6 and 4. Constant: -9.

8. In a soccer tournament a team earns 3 points for a win and 1 point for a tie. Define your variables, then write an expression for a team's total points. (Write)

Answer: Let w = wins and t = ties. Expression: $3w + t$

9. Using your expression from #8, find the points for a team with 5 wins and 2 ties. (Evaluate)

Answer: $3(5) + 2 = 17$ points

10. Error Analysis – Gus evaluated $2x^2$ when $x = 3$ and wrote 36. Identify his mistake, then evaluate correctly. (Error Analysis)

Answer: He multiplied $2 \cdot 3$ first and then squared: $(2 \cdot 3)^2 = 36$. The exponent belongs to x ONLY. Correct: $2(3^2) = 2(9) = 18$.

11. Write a verbal expression for $a^4 + 6b/7$, then write one for $(a^4 + 6b) \div 7$. Explain how the wording shows the difference. (Compare)

Answer: First: a number to the fourth power, plus the quotient of six times a second number and seven. Second: the SUM of a number to the fourth power and six times a second number, all divided by seven. The word "sum" placed first signals the grouping.

12. The surface area of a sphere is four times π times the radius squared. Write the expression, then find the surface area of a bubble with radius 12 cm. Round to the nearest tenth. (Write & Evaluate)

Answer: $4\pi r^2$; $4\pi(12)^2 = 4\pi(144) = 576\pi \approx 1809.6$ square centimeters